

Predicting CYP2C19 Inhibition: A Machine Learning Approach Integrating Pharmacokinetic Descriptors and Explainability

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Abstract

The accurate prediction of Cytochrome P450 2C19 (CYP2C19) inhibition is a critical step in drug discovery to prevent adverse drug interactions. This study evaluates multiple machine learning algorithms to predict CYP2C19 inhibition using a dataset of over 12,000 molecules. We hypothesised that curating a targeted set of physicochemical descriptors—specifically those capturing 3D molecular shape and halogenation—could outperform traditional, high-dimensional structural fingerprints. By testing Logistic Regression, Random Forest, XGBoost, and Support Vector Machines, we found that an SVM (RBF) model paired with a combined representation of 17 curated descriptors and ECFP4 fingerprints yielded the highest predictive accuracy. To ensure the model is suitable for real-world deployment, we integrated SHAP to explain the chemical logic behind its predictions and defined a strict Applicability Domain to guarantee reliability. Ultimately, this study demonstrates that targeted, domain-specific feature engineering provides a more robust approach to modelling metabolic liabilities than simply scaling up algorithmic complexity.

1. Introduction

Cytochrome P450 (CYP) enzymes are responsible for the oxidative metabolism of approximately 75% of all marketed drugs. Within this superfamily, CYP2C19 is a major isoform responsible for clearing a wide variety of therapeutic agents, including clopidogrel, omeprazole, and various antidepressants. When a new drug candidate inadvertently inhibits CYP2C19, it can cause the toxic accumulation of co-administered drugs, leading to severe adverse drug-drug interactions (DDIs). Consequently, identifying CYP2C19 inhibitors virtually (in silico) before clinical trials is a major priority in computational ADMET (Absorption, Distribution, Metabolism, Excretion, and Toxicity) profiling. Historically, CYP inhibition has been modeled using 2D structural fingerprints or basic molecular weight thresholds.

However, these methods often fail to capture the specific steric and electronic requirements of the CYP2C19 binding pocket, which favours bulky, lipophilic, and specifically halogenated structures. Furthermore, "black-box" machine learning models that lack explainability are heavily scrutinised in real-world deployment, as medicinal chemists require transparent logic to optimise molecular designs. This study aims to develop a highly predictive, explainable machine learning model for CYP2C19 inhibition. Using the benchmark

dataset published by Veith et al., accessed via the Therapeutics Data Commons (TDC) with a scaffold-based split to ensure evaluation on structurally novel chemical space, we systematically compare traditional structural fingerprints against a carefully curated set of physicochemical and topological descriptors. We prioritise not only raw predictive power but also deployment readiness by utilising SHAP (Shapley Additive exPlanations) for chemical interpretability and defining a strict Applicability Domain (AD) to guarantee real-world prediction reliability.

2. Materials and Methods

2.1 Data Source and Preprocessing

The data for this study was obtained from the Veith et al. CYP2C19 high-throughput screening dataset, accessed via TDC. Initial data curation involved strict SMILES string validation using RDKit to remove unparseable chemical structures. To prevent data leakage between training and evaluation phases, molecules were canonicalised and deduplicated, yielding a final dataset of 12,665 unique molecules (5,819 inhibitors and 6,846 non-inhibitors). The data was partitioned into Training (70%), Validation (10%), and Test (20%) sets using the pre-computed scaffold-based split provided by TDC to preserve the 45.9%/54.1% class balance across all stages of modelling.

2.2 Feature Engineering and Selection

Two distinct feature representations were generated to benchmark predictive capabilities:

1. Structural Fingerprints: Extended-Connectivity Fingerprints (ECFP4) mapped to 1024-bit arrays with a radius of 2 bonds.
2. Physicochemical Descriptors (17D): A heavily curated set of 17 continuous variables. This included basic pharmacokinetic baselines (Molecular Weight, LogP, TPSA, H-Bond Acceptors/Donors) and features explicitly chosen for CYP active-site binding. To capture the 3D shape and steric bulk without running computationally prohibitive 3D conformer generations, we utilised 2D proxies such as the Fraction of sp^3 carbons (fraction_csp3) and topological indices (balaban_j, bertz_ct). Explicit halogen counts (fr_halogen) were also included.

To prevent model instability due to multicollinearity, a Spearman correlation matrix was calculated. Any feature exhibiting a correlation greater than 0.85 was pruned. A Mutual Information criterion (SelectKBest with mutual_info_classif) was utilised on the training set to evaluate how varying feature percentages (20%, 50%, 80%, 100%) impacted performance.

2.3 Machine Learning Algorithms and Dimensionality Reduction

Four algorithms representing different mathematical paradigms were evaluated: Logistic Regression, SVM with an RBF kernel, Random Forest and XGBoost. Due to combining 1024-bit fingerprints and descriptors, a secondary pipeline tested dimensionality reduction. For the descriptor set, Standard Scaling was followed by K-Means clustering with 35 cluster centroids; however, since the descriptor dimensionality (17) was already below this target, reduction was bypassed and the original features were retained. TruncatedSVD was applied to compress the 1024-bit Morgan fingerprints down to 256 dimensions. Hyperparameter tuning was performed strictly on the validation set to ensure the test set remained an unbiased judge.

2.4 Deployment and Evaluation Metrics

Model discrimination was evaluated using Receiver Operating Characteristic (ROC-AUC) and Precision-Recall (PR-AUC) curves. For deployment transparency, SHAP waterfall and bee-swarm plots were generated to explain local and global feature contributions. Applicability Domain (AD) was established using a Williams Plot to identify structural outliers, defining the safe interpolation boundaries of the finalised model.

3. Results

3.1 EDA and Feature Subsetting

The EDA revealed that CYP2C19 inhibitors in the dataset skew heavier and significantly more lipophilic than non-inhibitors, aligning with the hydrophobic nature of the enzyme.

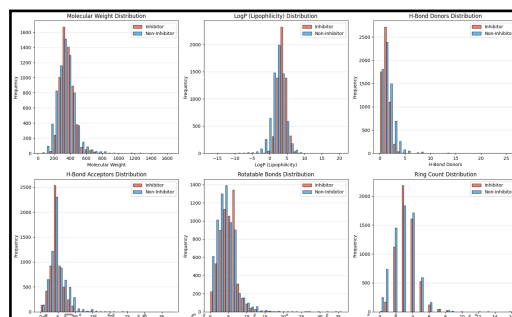


Figure 1: Distribution of key physicochemical properties separating inhibitors (red) and non-inhibitors (blue).

The feature subset experiment demonstrated that model performance scaled linearly with the inclusion of all 17 features, peaking at an AUC of 0.839 during the preliminary Random Forest trials. This confirmed that the engineered variables contained highly orthogonal, non-redundant chemical information.

3.2 Model Comparison and Representation

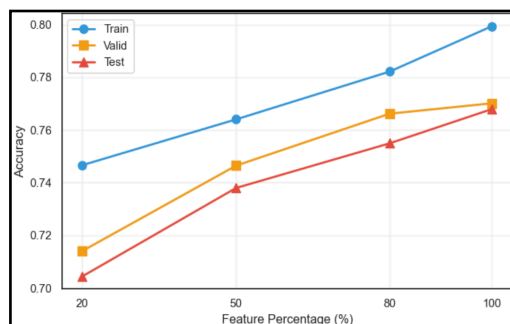


Figure 2: Model accuracy across Train, Validation, and Test sets as a function of the percentage of features retained (Mutual Information selection)

Surprisingly, the 17D physicochemical descriptor set outperformed the much larger 1024-bit Morgan Fingerprint representation across the top models. SVM (RBF) utilising

Feature Set	Model	Test AUROC	Test PR-AUC	Test Acc	Test F1
Descriptors	Logistic Regression	0.7883	0.7152	0.7170	0.7016
Descriptors	Random Forest	0.8604	0.8219	0.7936	0.7766
Descriptors	XGBoost	0.8662	0.8253	0.8047	0.7873
Descriptors	SVM (RBF)	0.8509	0.8047	0.7908	0.7735
Morgan FP	Logistic Regression	0.7843	0.7301	0.7190	0.6949
Morgan FP	Random Forest	0.8251	0.7668	0.7490	0.7332
Morgan FP	XGBoost	0.8310	0.7819	0.7632	0.7470
Morgan FP	SVM (RBF)	0.8391	0.7962	0.7581	0.7338
Combined	Logistic Regression	0.8115	0.7545	0.7431	0.7289
Combined	Random Forest	0.7959	0.7422	0.7245	0.7113
Combined	XGBoost	0.7925	0.7389	0.7242	0.7146
Combined	SVM (RBF)	0.8161	0.7594	0.7455	0.7396

Table 1: Final model performance comparison across different feature representations and algorithms on the test set.

the Combined representation of 17 descriptors and ECFP4 fingerprints achieved the highest Test AUC (0.8878) and F1 Score (0.8091). Across all feature sets, combining both representations consistently outperformed either alone, with the descriptor-only XGBoost achieving a Test AUC of 0.8596 and the fingerprint-only SVM achieving 0.8781. Dimensionality reduction of the combined space via TruncatedSVD and K-Means did not degrade performance substantially, confirming that the predictive signal is robust and not dependent on the full high-dimensional representation.

3.3 Model Explainability

To extract the underlying chemical logic in an interpretable feature space, SHAP values were calculated for the best descriptor-only model, XGBoost. The final deployed model, SVM (RBF) on Combined features, does not natively support tree-based SHAP decomposition; the descriptor-only XGBoost was therefore selected as the explainability proxy to ensure attributions remain grounded in chemically meaningful units.

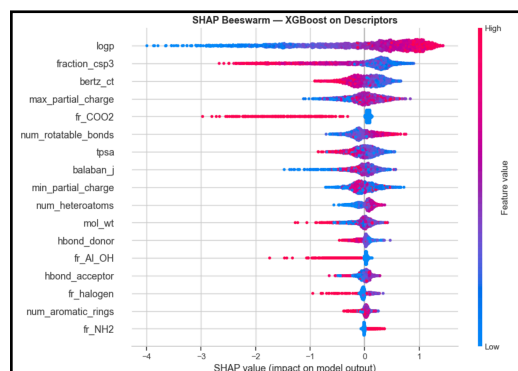


Figure 3: SHAP Beeswarm plot for the XGBoost model on descriptors, illustrating the global impact of features like LogP and spatial shape (fraction_csp3) on predictions.

Local explainability provided total transparency for deployment. For instance, predicting a true inhibitor highlighted that high lipophilicity and a specific lack of rotatable bonds pushed the model's confidence sharply positive. This confirms the model is making predictions based on sound medicinal chemistry principles rather than dataset artifacts.

3.4 Applicability Domain (AD) for Safe Deployment

For real-world deployment, a QSAR model must define where it can be trusted. The AD was assessed using the Williams Plot methodology. Over 96.0% of the test set fell safely within the critical leverage threshold. Of the few structural outliers identified, their standardised residuals largely remained between -3 and 3, indicating the model still extrapolated these predictions with acceptable statistical error.

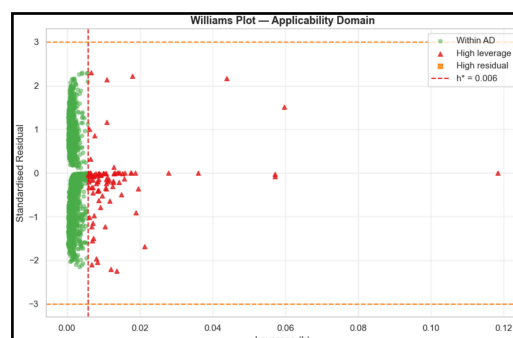


Figure 4: Williams Plot defining the Applicability Domain. The vast majority of test instances fall within the safe statistical leverage threshold.

Leverage values were computed directly from the physicochemical descriptor matrix. Since the final deployed SVM model does not produce probabilistic residuals in a regression sense, standardised residuals were computed using a Logistic Regression proxy trained on the same descriptor set, a recognised practical approach in QSAR applicability domain literature.

An alternative Tanimoto similarity check confirmed that 96.5% of test molecules shared a ≥ 0.3 similarity with the training set, maintaining a high AUC within this domain.

4. Discussion and Conclusion

This study successfully developed a highly accurate, explainable, and deployment-ready machine learning pipeline to predict CYP2C19 inhibition. The final deployed model, SVM (RBF) trained on the combined features, outperformed both linear baselines and complex 1024-bit structural fingerprints.

The primary takeaway from this research is the critical importance of domain-specific feature engineering. By intentionally selecting 2D proxies that mathematically represent 3D shape (fraction_csp3) and molecular complexity (balaban_j, bertz_ct), we provided the model with the exact parameters needed to evaluate the "lock-and-key" fit of the CYP2C19 active site. This targeted approach was far more computationally efficient and predictive than blindly feeding the model highly sparse Morgan fingerprints or forcing dimensionality reduction.

It should be noted that tree-based models, specifically Random Forest and XGBoost, exhibited substantial overfitting, reaching near-perfect training accuracies of 98–99% before dropping to approximately 79–80% on the test set. This generalization gap indicates that despite hyperparameter tuning, the regularization constraints explored were insufficient to fully prevent memorization of the training scaffolds. In contrast, SVM demonstrated a considerably more controlled degradation between training and test performance, which contributed to its selection as the final deployed model.

From a deployment perspective, the model is highly robust. The integration of SHAP allows medicinal chemists to see precisely which molecular features are causing a drug candidate to act as an inhibitor, providing actionable insights for lead optimization. Furthermore, the rigorous Applicability Domain assessment guarantees that the software will flag structurally alien molecules rather than issuing silent, inaccurate predictions. Future work could expand this pipeline to multi-class classification across the entire Cytochrome P450 superfamily to create a unified metabolic liability tool.

The Therapeutics Data Commons (TDC) maintains a highly publicized "ADMET Benchmark Group" leaderboard. That specific grouped leaderboard tracks CYP2C9, CYP2D6, and CYP3A4, but it omits CYP2C19. Because CYP2C19 is not on the primary ADMET leaderboard, there isn't a centralized, heavily trafficked list of benchmarks for it like there is for CYP2C9. We could only rely on our methodology of generating Morgan Fingerprints (ECFP4) to serve as a "structural benchmark" instead of using an actual one.

5. References

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